

Optimal Execution Under Stochastic Volatility: From Almgren–Chriss to the Heston Framework

Nathan Hoehndorf, Milo Glennon, Ben Guevel, Nathan Hess, Danielle Spahr

May 13th, 2026

A Derivation of Dynamic Trading Strategies via Hamilton–Jacobi–Bellman Theory, Asymptotic Expansion, and Monte Carlo Empirical Analysis

Abstract

We study the problem of optimal portfolio liquidation under market impact. Beginning with the classical Almgren–Chriss (AC) framework [almgren2001optimal], we derive the static optimal trading trajectory via a discrete calculus-of-variations approach. We then identify the principal limitation of the AC model—its assumption of constant volatility—and extend the framework by coupling the asset price dynamics to a Heston stochastic volatility process [heston1993closed]. The resulting control problem is analyzed through the Hamilton–Jacobi–Bellman (HJB) partial differential equation [fleming2006controlled]. Because the HJB equation has no closed-form solution in the fully coupled system, we apply an asymptotic perturbation expansion in the volatility-of-volatility parameter ξ [fouque2011multiscale], recovering an analytically tractable, volatility-adjusted optimal trading rate. We conclude with a description of the Monte Carlo simulation architecture [glasserman2004monte] used to compare the two frameworks empirically, including full statistical testing and calibration against LOBSTER limit-order-book data [huang2011lobster].

Contents

1	Introduction and the Fundamental Trade-off	2
2	The Static Almgren–Chriss Framework	2
2.1	System Dynamics	2
2.2	Implementation Shortfall	3
2.3	Expected Implementation Shortfall	3
2.4	Variance of Implementation Shortfall	4
2.5	The Mean–Variance Objective and Euler–Lagrange Conditions	4
2.6	The Optimal Static Trajectory	4

3	Introducing Stochastic Volatility: A Four-State Formulation	5
3.1	Motivation	5
3.2	The Controlled Heston Dynamics	5
4	Dynamic Optimization and the HJB Equation	6
4.1	The Dynamic Value Function	6
4.2	Bellman’s Principle and the Itô Expansion	6
4.3	The Four-State HJB Equation	7
5	Quadratic Ansatz and Asymptotic Perturbation Expansion	7
5.1	Reduction via the Mark-to-Market Decomposition	7
5.2	Zeroth-Order Equation	7
5.3	Recovery of the Almgren–Chriss Solution	7
6	First-Order Correction and the Feynman–Kac Representation	8
6.1	First-Order PDE	8
6.2	Feynman–Kac Representation	8
7	Volatility-Adjusted Optimal Trading Rate	8
7.1	The Correction Coefficient h_v	9
8	Computational Architecture	9
8.1	Overview	9
8.2	Parameter Calibration from LOBSTER Data	9
8.3	Monte Carlo Simulation	10
8.4	Statistical Test Suite	10
9	Dataset Sweep and Cross-Dataset Validation	10
10	Empirical Results	11
10.1	Stage 1: Theoretical Verification via Synthetic Control	11
10.2	Stage 2: Empirical Application to LOBSTER Data	12
10.3	Stage 3: Discrepancy Diagnosis	13
A	Full Derivations of the IS Statistics	15
A.1	Expected Value of Implementation Shortfall	15
A.2	Variance of Implementation Shortfall	15
B	Ansatz Decomposition: Financial and Mathematical Justification	15
B.1	Financial Intuition	15
B.2	Mathematical Justification via the Five-State System	16
B.3	Brownian Correlation via Cholesky	16

C	Euler–Maruyama Discretization and Numerical Stability	16
C.1	Full Discretization Scheme	16
C.2	Feller Condition and Full Truncation	16
C.3	Path Filtering	16
D	Cholesky Decomposition for Correlated Shocks	17
E	Riccati ODE: Full Derivation and Solution	17
F	Parameter Calibration from LOBSTER Data	17
F.1	Data Format	17
F.2	Volatility and Heston Parameters	17
F.3	Market Impact Parameters	18
G	Implementation Shortfall Decomposition	18
H	Monte Carlo Simulation Design	18
H.1	Paired Architecture	18
H.2	Variance Reduction	18
H.3	Sample Size	18
I	Volatility Regime Partitioning	19
I.1	Regime Dependence and Model Risk	19
J	Statistical Tests: Full Definitions and Hypotheses	19
J.1	Paired t -Test	19
J.2	Wilcoxon Signed-Rank Test	20
J.3	Levene’s Test for Equality of Variance	20
J.4	Two-Sample KS Test	20
J.5	CVaR via Bootstrap	20
J.6	Sharpe-Like Reliability Ratio	20
K	Implementation Details and Engineering Decisions	20
K.1	Time-Scale Convention	20
K.2	Exact AC Baseline and Corrected Trade Generation	20
K.3	Quadrature for $b(t)$	21
K.4	Parallelisation	21
K.5	Random Number Generation	21
K.6	Paired Comparison Design	21

1. Introduction and the Fundamental Trade-off

Consider a trader who begins the day holding a large inventory X of a single asset and must fully liquidate it by a fixed deadline T . Selling the entire position at once is almost never feasible: the market’s limit order book (LOB) is a snapshot of liquidity, showing all bids and asks organized by price. The LOB can only absorb so many shares at the prevailing bid price before prices begin to deteriorate. The trader is therefore forced to break the liquidation into smaller tranches executed over time.

This creates a fundamental tension. Trading *slowly* reduces the cost of market impact—each slice is small enough that the book absorbs it with minimal price depression—but exposes the remaining inventory to prolonged market risk: the asset price may fall while shares are still held. Trading *quickly* eliminates that exposure but hammers the price, extracting poor execution. The goal of optimal execution theory is to find the trajectory $\{n_t\}_{0 \leq t \leq T}$ that navigates this trade-off optimally [kyle1985continuous].

The Almgren–Chriss (AC) model, introduced in a seminal 2001 paper by Robert Almgren and Neil Chriss [almgren2001optimal], was the first to provide a tractable analytical solution to this problem using a mean-variance objective. It remains the industry standard starting point and serves as the baseline for the more sophisticated model developed here.

The primary deficiency of the AC framework is its assumption that asset volatility σ^2 is a fixed constant throughout the entire liquidation window. Real volatility is anything but constant: it spikes on earnings announcements, mean-reverts during calm periods, and is itself a stochastic process [engle1982arch]. A static, pre-planned schedule derived under the assumption of constant volatility cannot optimally adapt to this reality. The Heston stochastic volatility model captures these dynamics [heston1993closed], which is precisely what the present work develops.

2. The Static Almgren–Chriss Framework

2.1 System Dynamics

We model the liquidation of X shares over a horizon $[0, T]$ subdivided into N intervals of length $\tau = T/N$. Let x_k denote the inventory at the start of period k , so $x_0 = X$ and $x_N = 0$. At each step, the trader submits an order of size

$$n_k = x_{k-1} - x_k.$$

The trading velocity (shares per unit time) is

$$v_k = \frac{n_k}{\tau}.$$

The fundamental mid-price evolves as

$$S_k = S_{k-1} - \gamma v_k \tau + \sigma \sqrt{\tau} \varepsilon_k,$$

where $\varepsilon_k \stackrel{\text{iid}}{\sim} N(0, 1)$, $\sigma > 0$ is the asset’s daily volatility, and $\gamma > 0$ is the permanent market impact

parameter.

The realized execution price includes an additional temporary impact penalty:

$$\tilde{S}_k = S_{k-1} - \eta v_k,$$

where $\eta > 0$ is the temporary impact parameter.

Remark 2.1 (Two Types of Market Impact). Permanent impact *represents a shift in market belief about fair value caused by the information content or size of the trade*. Temporary impact *is the transient cost of consuming available liquidity; once the trade is complete and the order book refills, the price reverts* [kyle1985continuous].

2.2 Implementation Shortfall

The implementation shortfall (IS) [perold1988implementation] is defined as

$$\text{IS} = XS_0 - \sum_{k=1}^N n_k \tilde{S}_k.$$

2.3 Expected Implementation Shortfall

Proposition 2.2 (Expected IS). *Under the linear impact model,*

$$\mathbb{E}[\text{IS}] = \sum_{k=1}^N \frac{\eta n_k^2}{\tau} + \frac{1}{2} \gamma X^2.$$

Proof. Substitute the capture price into the IS definition:

$$\text{IS} = XS_0 - \sum_{k=1}^N n_k \tilde{S}_k = XS_0 - \sum_{k=1}^N n_k S_{k-1} + \eta \sum_{k=1}^N \frac{n_k^2}{\tau}.$$

Expand the mid-price telescopically:

$$S_{k-1} = S_0 + \sum_{j=1}^{k-1} (\sigma \sqrt{\tau} \varepsilon_j - \gamma n_j).$$

After substitution and using $\mathbb{E}[\varepsilon_j] = 0$,

$$\mathbb{E}[\text{IS}] = \gamma \sum_{k=1}^N n_k \sum_{j=1}^{k-1} n_j + \eta \sum_{k=1}^N \frac{n_k^2}{\tau}.$$

The permanent impact double sum telescopes via $(\sum n_k)^2 = \sum n_k^2 + 2 \sum_k n_k \sum_{j < k} n_j = X^2$ to give $\frac{1}{2} \gamma (X^2 - \sum n_k^2)$. Combining and taking the continuum limit ($\tau \rightarrow 0$, so the $-\frac{1}{2} \gamma \sum n_k^2$ term vanishes) yields the result. $\square$

2.4 Variance of Implementation Shortfall

Proposition 2.3 (Variance of IS).

$$\text{Var}[\text{IS}] = \sigma^2 \sum_{k=1}^N \tau x_k^2.$$

Proof. Only the stochastic component contributes to variance:

$$Z = -\sigma\sqrt{\tau} \sum_{j=1}^N x_{j-1}\varepsilon_j,$$

where we have used the Abel summation identity $\sum_{k=1}^N n_k \sum_{j=1}^{k-1} \varepsilon_j = \sum_{j=1}^{N-1} x_j \varepsilon_j$. Using independence of the ε_j ,

$$\text{Var}[\text{IS}] = \sigma^2 \tau \sum_{j=1}^N x_{j-1}^2. \quad \square$$

2.5 The Mean–Variance Objective and Euler–Lagrange Conditions

A trader with risk-aversion parameter $\lambda > 0$ minimizes

$$U = \mathbb{E}[\text{IS}] + \lambda \text{Var}[\text{IS}] = \sum_{k=1}^N \left(\frac{\eta n_k^2}{\tau} + \lambda \sigma^2 \tau x_k^2 \right) + \text{const.}$$

The Euler–Lagrange condition for this discrete variational problem is

$$\eta(x_{k+1} - 2x_k + x_{k-1}) = \lambda \sigma^2 \tau^2 x_k.$$

Remark 2.4 (A Discrete Equation of Motion). *The left-hand side is the discrete analogue of the second derivative of the inventory trajectory; the right-hand side is the restoring force proportional to inventory.*

2.6 The Optimal Static Trajectory

Theorem 2.5 (Almgren–Chriss Optimal Inventory [almgren2001optimal]). *Define*

$$\kappa = \sqrt{\frac{\lambda \sigma^2}{\eta}}.$$

The unique solution satisfying $x_0 = X$ and $x_N = 0$ is

$$x_k = \frac{\sinh(\kappa(T - t_k))}{\sinh(\kappa T)} X.$$

Proof. Rewrite the Euler–Lagrange equation as

$$x_{k+1} - (2 + \kappa^2 \tau^2) x_k + x_{k-1} = 0.$$

Using the ansatz $x_k = r^k$ yields the characteristic equation $r^2 - (2 + \kappa^2 \tau^2)r + 1 = 0$. The two roots satisfy $r_1 r_2 = 1$, so the general solution is a linear combination of the hyperbolic sine and cosine centered at κ . Imposing $x_0 = X$ and $x_N = 0$ selects

$$x_k = \frac{\sinh(\kappa(T - t_k))}{\sinh(\kappa T)} X. \quad \square$$

The continuous-time optimal trading rate is

$$n_t^{(\text{AC})} = -\dot{x}_t = \kappa \frac{\cosh(\kappa(T - t))}{\sinh(\kappa T)} X = \kappa \coth(\kappa(T - t)) x_t.$$

3. Introducing Stochastic Volatility: A Four-State Formulation

3.1 Motivation

The classical Almgren–Chriss framework assumes constant volatility and therefore produces a deterministic liquidation trajectory fixed at time $t = 0$. Real markets exhibit stochastic volatility clustering, leverage effects, and regime-dependent execution risk [engle1982arch, black1976studies]. To incorporate these effects dynamically, we extend the state space to include both the underlying asset price and a stochastic variance process. The resulting control problem now evolves over the four-dimensional state vector (t, x, S, v) , where t is time, x_t is remaining inventory, S_t is the mid-price, and v_t is instantaneous variance.

3.2 The Controlled Heston Dynamics

Definition 3.1 (Four-State Controlled System). *The controlled dynamics are:*

$$dx_t = -n_t dt, \quad dS_t = -\gamma n_t dt + \sqrt{v_t} dW_t^S, \quad dv_t = \theta(\omega - v_t)dt + \xi \sqrt{v_t} dW_t^v,$$

with $dW_t^S dW_t^v = \rho dt$.

The execution price remains $\tilde{S}_t = S_t - \eta n_t$.

Symbol	Name	Interpretation
X	Initial inventory	Shares to liquidate
T	Trading horizon	Final liquidation deadline
γ	Permanent impact	Long-run price depression per share
η	Temporary impact	Instantaneous liquidity cost
λ	Risk aversion	Inventory risk penalty
ω	Long-run variance	Mean reversion level
θ	Mean reversion speed	Variance pullback intensity
ξ	Vol-of-vol	Variance diffusion strength
ρ	Leverage effect	Price–volatility correlation

4. Dynamic Optimization and the HJB Equation

4.1 The Dynamic Value Function

The trader minimizes expected total remaining execution cost plus inventory risk [oksendal2003sde]:

$$J(t, x, S, v) = \inf_{\{n_s\}_{s \geq t}} \mathbb{E}_{t,x,S,v} \left[\int_t^T (\eta n_s^2 + \lambda v_s x_s^2) ds - x_T S_T \right].$$

Because the asset price enters linearly through the liquidation value, it is natural to separate the mark-to-market contribution from the nonlinear control component via the decomposition

$$J(t, x, S, v) = -xS + \Phi(t, x, v).$$

The term $-xS$ is the mark-to-market value of the remaining inventory; Φ captures the nonlinear execution-risk correction. This decomposition is justified rigorously via a five-dimensional wealth formulation (see Appendix B), which shows that the spot price S cancels completely from the HJB control terms.

4.2 Bellman’s Principle and the Itô Expansion

Bellman’s principle [fleming2006controlled] states

$$0 = \inf_n \left\{ \eta n^2 dt + \lambda v x^2 dt + \mathbb{E}[dJ] \right\}.$$

Applying multivariate Itô calculus [karatzas1998brownian] and using $(dS_t)^2 = v_t dt$, $(dv_t)^2 = \xi^2 v_t dt$, $dS_t dv_t = \rho \xi v_t dt$, and $(dx_t)^2 = dx_t dS_t = dx_t dv_t = 0$, the Hamiltonian becomes

$$\mathcal{H}(n) = \eta n^2 - n(J_x + \gamma J_S),$$

with first-order condition $n^* = \frac{1}{2\eta}(J_x + \gamma J_S)$.

4.3 The Four-State HJB Equation

Theorem 4.1 (Four-State HJB Equation [fleming2006controlled]). *The value function satisfies*

$$0 = J_t + \lambda v x^2 + \theta(\omega - v)J_v + \frac{1}{2}vJ_{SS} + \frac{1}{2}\xi^2 vJ_{vv} + \rho\xi vJ_{Sv} - \frac{1}{4\eta}(J_x + \gamma J_S)^2,$$

with terminal condition $J(T, x, S, v) = -xS$.

5. Quadratic Ansatz and Asymptotic Perturbation Expansion

5.1 Reduction via the Mark-to-Market Decomposition

Using $J = -xS + \Phi(t, x, v)$ [fouque2011multiscale], the second-order S -derivatives vanish ($J_{SS} = J_{Sv} = 0$) and the HJB reduces to

$$0 = \Phi_t + \lambda v x^2 + \theta(\omega - v)\Phi_v + \frac{1}{2}\xi^2 v\Phi_{vv} - \frac{1}{4\eta}(\Phi_x - \gamma x)^2.$$

We expand in powers of the volatility-of-volatility parameter ξ :

$$\Phi = \Phi^{(0)} + \xi\Phi^{(1)} + O(\xi^2).$$

5.2 Zeroth-Order Equation

At order $O(1)$, adopting the quadratic ansatz $\Phi^{(0)} = \frac{1}{2}\gamma x^2 + h(t, v)x^2$ (which is valid because the differential operators preserve polynomial degree in x), we obtain $\Phi_x^{(0)} - \gamma x = 2h(t, v)x$ and the equation collapses to the nonlinear Riccati PDE

$$0 = h_t + \theta(\omega - v)h_v + \lambda v - \frac{h^2}{\eta}.$$

5.3 Recovery of the Almgren–Chriss Solution

When $v_t \equiv \sigma^2$ (constant), $h_v = 0$ and the PDE reduces to the classical Riccati ODE (see Appendix E for the full derivation):

$$h_t + \lambda\sigma^2 - \frac{h^2}{\eta} = 0, \quad h(T) = +\infty,$$

with solution $h(t) = \eta\kappa \coth(\kappa(T - t))$. The optimal control then gives precisely the AC rate:

$$n_t^{(0)} = \kappa \coth(\kappa(T - t)) x_t.$$

6. First-Order Correction and the Feynman–Kac Representation

6.1 First-Order PDE

Collecting $O(\xi)$ terms, the effective generator on the variance component is

$$\mathcal{L}^{(v)} f = \theta(\omega - v) f_v + \frac{1}{2} \xi^2 v f_{vv},$$

and the correction satisfies the compact equation

$$\left(\partial_t + \mathcal{L}^{(v)} \right) \Phi^{(1)} - \frac{h}{\eta} x \Phi_x^{(1)} = -\rho v x h_v.$$

6.2 Feynman–Kac Representation

Since this PDE is linear in $\Phi^{(1)}$, the Feynman–Kac theorem [karatzas1998brownian] gives a stochastic representation: along the zeroth-order characteristic $dx_s = -\frac{h(s, v_s)}{\eta} x_s ds$,

$$\Phi^{(1)}(t, x, v) = \mathbb{E}_{t, x, v} \left[\int_t^T \rho v_s x_s h_v(s, v_s) ds \right].$$

Using the frozen-variance approximation $v_s \approx v_t$ and the AC inventory $x_s = x_t \sinh(\kappa(T - s)) / \sinh(\kappa(T - t))$, the integral evaluates to $\frac{1}{\kappa} \tanh\left(\frac{\kappa(T - t)}{2}\right)$, giving

$$\Phi^{(1)} \approx \frac{\rho v_t x_t h_v}{\kappa} \tanh\left(\frac{\kappa(T - t)}{2}\right).$$

7. Volatility-Adjusted Optimal Trading Rate

Differentiating the full value function $J = J^{(0)} + \xi J^{(1)} + O(\xi^2)$ and substituting into $n^* = \frac{1}{2\eta}(J_x + \gamma J_s)$ gives:

$$n_t^* \approx \kappa \coth(\kappa(T - t)) x_t - \frac{\xi \rho v_t h_v}{2\eta \kappa} \tanh\left(\frac{\kappa(T - t)}{2}\right).$$

Remark 7.1 (Interpretation). *The first term is the classical AC liquidation speed. The second term adjusts dynamically:*

- $\rho < 0$ (equity leverage effect): rising variance slows liquidation.
- $\rho > 0$: rising variance accelerates liquidation.
- $\rho = 0$: the correction vanishes and the rate reduces to pure AC.

The correction is strongest mid-horizon where inventory and variance uncertainty jointly dominate.

7.1 The Correction Coefficient h_v

The derivative $h_v = \partial h / \partial v$ is approximated via a secondary perturbation expansion. Setting $h(t, v) = h^{(0)}(t) + \xi(a(t) + b(t)v) + O(\xi^2)$, so $h_v \approx \xi b(t)$, and substituting into the Riccati PDE yields the linear ODE **[CITATION NEEDED: standard technique for affine Heston Riccati perturbations]**

$$\dot{b}(t) - \left(\theta + \frac{2h^{(0)}(t)}{\eta} \right) b(t) = -\frac{\lambda}{\xi},$$

which, upon integration with the integrating factor $\mu(t) = e^{-\theta t} \sinh^2(\kappa(T-t))$ and terminal condition $b(T) = 0$, gives:

$$b(t) = \frac{\lambda}{\xi} \int_t^T e^{-\theta(s-t)} \left(\frac{\sinh(\kappa(T-s))}{\sinh(\kappa(T-t))} \right)^2 ds.$$

The perturbation ansatz $h^{(1)} = a(t) + b(t)v$ is standard for affine Heston-type models **[CITATION NEEDED: affine structure of Heston model, e.g. Duffie–Pan–Singleton (2000)]**; the specific form here follows from the affine-in- v structure of the zeroth-order solution.

8. Computational Architecture

8.1 Overview

The framework investigates whether stochastic-volatility market dynamics improve execution outcomes relative to the classical AC model. The full implementation is available at the project repository. All parameters are expressed in units of *one trading day* ($T = 1$), so that σ is daily volatility, v_0 and ω are daily variance, and $N = 78$ intervals correspond to one 5-minute bucket per interval over a 6.5-hour trading day.

8.2 Parameter Calibration from LOBSTER Data

Empirical parameters are extracted from LOBSTER limit-order-book data **[huang2011lobster]**. The calibration pipeline is described fully in Appendix F; key steps include:

1. Mid-price log returns at 5-minute and 1-minute frequencies.
2. Daily volatility σ via square-root-of-time scaling **[CITATION NEEDED: standard volatility scaling, e.g. Andersen–Bollerslev (1997)]**.
3. Temporary impact η by walking the limit order book at simulated trade sizes.
4. Permanent impact γ by regressing 5-minute lagged signed order flow against future mid-price changes.
5. Heston parameters $(\theta, \omega, \xi, \rho)$ from an OLS fit to the first-difference dynamics of rolling 1-minute realized variance, scaled to daily units.

The Feller condition $2\theta\omega \geq \xi^2$ **[cox1985theory]** is enforced after calibration by reducing ξ if necessary.

8.3 Monte Carlo Simulation

For each simulation i [glasserman2004monte]:

- (i) Draw one shared Heston path $(S_k^{(i)}, v_k^{(i)})_{k=0}^N$ using the log-Euler full-truncation scheme (Appendix C).
- (ii) Evaluate the **AC strategy**: compute the static schedule n_k^{AC} from the initial v_0 and apply it to the realized path.
- (iii) Evaluate the **Heston strategy**: at each step, recompute n_k^* using the current realized v_k ; apply to the same path.
- (iv) Record $IS_{\text{AC}}^{(i)}$ and $IS_{\text{Heston}}^{(i)}$ for paired analysis.

The paired design ensures $IS_{\text{AC}}^{(i)} - IS_{\text{Heston}}^{(i)}$ reflects only the scheduling difference, not random variation in market conditions.

8.4 Statistical Test Suite

The empirical comparison applies seven statistical procedures, described fully in Appendix J:

- Paired t -test [lehmann2005testing]: tests $\mathbb{E}[IS_{\text{AC}} - IS_{\text{Heston}}] > 0$.
- Wilcoxon signed-rank test [lehmann2005testing]: nonparametric median test.
- Levene variance test [levene1960robust, brown1974robust]: tests $\text{Var}[IS_{\text{AC}}] > \text{Var}[IS_{\text{Heston}}]$.
- Two-sample KS test [kolmogorov1933sulla, smirnov1948table]: detects distributional differences.
- CVaR at 95% and 99% [rockafellar2000cvar] with bootstrap CIs [efron1993bootstrap].
- Volatility-regime analysis: within-tercile Wilcoxon tests.
- Leverage-effect sweep: Spearman correlation of ρ against mean IS difference [CITATION NEEDED: Spearman rank correlation, e.g. Spearman (1904)].

9. Dataset Sweep and Cross-Dataset Validation

The framework supports automated evaluation across all available LOBSTER datasets [huang2011lobster]. For each dataset, the calibration and paired Monte Carlo pipeline runs independently. Cross-dataset paired t -tests then test whether the Heston strategy is systematically superior across diverse market environments (Apple, Amazon, Google, Intel, Microsoft; various orderbook-depth configurations; June 21, 2012).

A Sharpe-like reliability ratio [CITATION NEEDED: Sharpe ratio analogue for execution costs; see e.g. Almgren and Lorenz (2006)]

$$\mathcal{R}_{j,d} = \frac{\overline{IS}_{j,d}}{s_{IS_{j,d}}}$$

is also compared across strategies and datasets, where a less negative value indicates more consistent execution.

10. Empirical Results

We organise the empirical evidence in three stages. Section 10.1 establishes a *synthetic control* that confirms the correction works correctly when the model’s assumptions are satisfied. Section 10.2 applies the same framework to real market data and finds no measurable advantage. Section 10.3 identifies, via side-by-side parameter comparison and path-level diagnostics, the precise mechanisms that suppress the correction in practice.

10.1 Stage 1: Theoretical Verification via Synthetic Control

Before testing on real data we validate the implementation in an environment that strictly obeys the model’s assumptions: a pure Heston price-variance system with no microstructure noise, no calibration error, and parameters set to produce economically significant variance dynamics.

Synthetic environment design

We fix the execution parameters as in Table 1 and sweep the

Parameter	Value	Description
S_0	100.0	Initial price
X	10,000	Shares to liquidate
T	1.0	Horizon (one trading day)
N	78	Intervals (one per 5 minutes)
σ	0.08	Daily volatility
λ	0.5	Risk aversion
η	0.01	Temporary impact coefficient
γ	10^{-6}	Permanent impact coefficient

Table 1: Fixed execution parameters for the synthetic control sweep.

three Heston parameters that most directly control the correction signal:

Parameter	Values swept	Economic meaning
ρ	$\{-0.7, -0.5, -0.3, 0.0\}$	Leverage effect strength
ξ	$\{0.10, 0.30\}$	Stochasticity of variance
v_0/ω	$\{1\times, 2\times, 4\times\}$	Magnitude of initial variance shock

The long-run variance is fixed at $\omega = 0.04$ (daily vol $\approx 20\%$) and the mean-reversion speed at $\theta = 2.0$ (half-life 135 minutes). Scenarios with $\xi^2 > 2\theta\omega$ are excluded as Feller-violating. All 18 valid scenarios use $n_{\text{sims}} = 1,000$ paired Heston paths.

Results

In all 18 valid scenarios the Heston correction reduces expected IS significantly ($p < 10^{-4}$ by both paired t -test and Wilcoxon, verified). The improvement scales monotonically with the strength of each driving parameter:

Factor	Range	Mean IS improvement	Qualitative direction
ρ	$-0.7 \rightarrow -0.3$	\$711 $\rightarrow$ \$335	Proportional to $ \rho $
ξ	$0.10 \rightarrow 0.30$	\$209 $\rightarrow$ \$581	Grows with ξ^2
Shock	$1 \times \rightarrow 4 \times \omega$	\$201 $\rightarrow$ \$628	Grows with $v_0 - \omega$

The best single scenario ($\rho = -0.7$, $\xi = 0.30$, shock = $4 \times$) achieves $\bar{d} = \$1,595$, $t = 25.7$, $p \approx 0$. The $\rho = 0$ scenarios produce $\bar{d} \equiv 0$ exactly, confirming the correction term vanishes when the leverage effect is absent.

Remark 10.1. *These results constitute a proof of concept: the implementation is correct and the mathematical structure of the correction is sound. The observed gains are large, monotone, and structurally explained by the perturbation formula.*

10.2 Stage 2: Empirical Application to LOBSTER Data

We now apply the identical framework to calibrated LOBSTER limit-order-book data for five US equities on June 21, 2012.

Per-Dataset Results

Dataset	Mean IS (AC)	Mean IS (Heston)	Diff (AC–Heston)
AAPL 1-level	50,420.98	50,420.98	+0.00
AAPL 10-level	12,902.51	12,902.51	+0.01
AAPL 5-level	20,289.78	20,289.77	+0.01
AMZN 1-level	48,106.06	48,106.07	−0.01
AMZN 10-level	9,828.15	9,828.20	−0.05
AMZN 5-level	18,017.72	18,017.75	−0.03
GOOG 1-level	106,911.38	106,911.38	−0.00
GOOG 10-level	41,151.37	41,151.37	−0.00
GOOG 5-level	60,463.06	60,463.06	−0.00
INTC 10-level	238.80	238.82	−0.02
INTC 5-level (a)	213.19	213.22	−0.03
INTC 5-level (b)	269.30	269.33	−0.03
MSFT 1-level	384.57	384.57	−0.00
MSFT 10-level	229.74	229.74	−0.00
MSFT 5-level	255.80	255.80	−0.00

Statistical tests (7,500 pooled pairs)

Test	Statistic	p -value	Reject H_0 ?
Paired t -test	$t = -9.13$	1.000 (one-sided)	No
Wilcoxon signed-rank	$W^+ = 12,622,962$	1.000	No
Levene variance	$F < 0.0001$	0.500	No
Two-sample KS	$D = 0.0004$	1.000	No

Across-dataset test

Quantity	Value
Mean diff AC–Heston (across datasets)	−0.011
t -statistic	−2.615
p -value (one-sided)	0.990
Sharpe-like ratio AC / Heston	−1.11 / −1.11

Volatility-regime analysis

Regime	Vol. range	n	Median diff. (AC–Heston)	p -value
Low	[0.006, 0.009]	2500	+0.00	0.0004 (Yes)
Mid	[0.009, 0.012]	2500	−0.00	1.000 (No)
High	[0.012, 0.016]	2500	−0.01	1.000 (No)

Finding: The Heston strategy is statistically indistinguishable from AC on all tests. IS differences are fractions of a cent regardless of volatility regime.

10.3 Stage 3: Discrepancy Diagnosis

The contradiction between Stage 1 (strong outperformance) and Stage 2 (zero measurable effect) is not a modelling error; it is a consequence of a systematic mismatch between the parameter regime assumed in the theory and the parameter regime produced by LOBSTER calibration. We isolate three mechanisms.

3.1 The vol-of-vol gap

Parameter	Synthetic (best scenario)	LOBSTER (mean calibrated)
ξ	0.30	0.0612
ω	0.04	0.00013
θ	2.0	39.15
v_0	0.08	0.00014
ρ	−0.50	−0.0097

The correction term scales as $O(\xi^2 v_t |\rho|)$. In the best synthetic scenario this product equals approximately $0.30^2 \times 0.08 \times 0.50 = 3.6 \times 10^{-3}$. In the LOBSTER environment it equals $0.06^2 \times 0.00014 \times 0.010 = 5 \times 10^{-9}$. The suppression factor is approximately **700,000** $\times$.

3.2 Quadratic variation of the variance path

Total quadratic variation (TQV) of v_t over $[0, T]$ measures how much the variance path actually moves during the execution window; it is the key signal that the correction can exploit.

Quantity	Synthetic	LOBSTER
Mean TQV = $\mathbb{E}\left[\sum_{k=1}^N (\Delta v_k)^2\right]$	5.4×10^{-3}	6.1×10^{-7}
Mean $\max(v_t) - \min(v_t)$ (path range)	0.0908	0.000406
Mean $ v_T - v_0 $ (net displacement)	0.0418	0.0000734

The LOBSTER variance paths are **8,700** $\times$ less variable than the synthetic paths by TQV. Under the LOBSTER calibration, the variance process has essentially no exploitable variation over the execution horizon: the high mean-reversion speed ($\theta \approx 39$, half-life < 7 minutes) absorbs every shock before the next 5-minute trading interval.

3.3 Mean-reversion speed and the effective half-life

Quantity	Formula	Synthetic	LOBSTER
Half-life	$\ln 2 / \theta$	0.35 days (135 min)	0.018 days (6.9 min)
Intervals per half-life	$(\ln 2 / \theta) / \tau$, $\tau = 1/78$	27	1.1

With $\theta \approx 39$ and $N = 78$ intervals of length $\tau = 1/78$ day, variance reverts to its mean in roughly *one interval*. The correction, which integrates the expected future variance trajectory via $b(t)$, therefore receives a signal that decays to near-zero within a single time step. The Heston correction was derived for slow mean reversion ($\theta \sim O(1)$) relative to the trading frequency; at $\theta = 39$ the correction operates in a regime outside the perturbation's regime of validity.

3.4 Near-zero leverage

The LOBSTER-calibrated ρ averages -0.010 with standard deviation 0.12 . The leverage parameter enters the correction linearly; at $|\rho| < 0.01$ the correction is numerically negligible regardless of ξ or v_t . Table 10.3 shows the synthetic correction at $\rho = 0$ is exactly zero by construction; the LOBSTER ρ is indistinguishable from zero at the intraday frequency on these data.

3.5 Side-by-side IS comparison under matched paths

Environment	$\bar{d}$ (AC-Heston)	t	p (one-sided)	$\max \Delta n_k $
Synthetic ($\rho = -0.7$, $\xi = 0.30$, shock=4 $\times$)	+1,595	25.7	0.000	8.11
LOBSTER (pooled, 15 datasets)	-0.01	-9.1	1.000	< 0.001

Summary of diagnosis. The three suppression mechanisms act multiplicatively: a $5\times$ reduction from smaller $|\rho|$, a $25\times$ reduction from smaller ξ , a $300\times$ reduction from smaller v_t , and a $\sim 100\times$ reduction from near-instantaneous mean reversion (which collapses the $b(t)$ integral). Together they account for the $\sim 700,000\times$ reduction in correction amplitude from the synthetic to the LOBSTER setting. The implementation is correct; the theory does not predict benefit in a market regime characterized by fast mean-reverting, low-amplitude intraday variance with negligible leverage correlation.

A. Full Derivations of the IS Statistics

A.1 Expected Value of Implementation Shortfall

Starting from $S_{k-1} = S_0 - \gamma \sum_{j=1}^{k-1} n_j + \sigma\sqrt{\tau} \sum_{j=1}^{k-1} \varepsilon_j$:

$$\begin{aligned} IS &= X S_0 - \sum_{k=1}^N n_k (S_{k-1} - \eta v_k) \\ &= \gamma \sum_{k=1}^N n_k \sum_{j=1}^{k-1} n_j - \sigma\sqrt{\tau} \sum_{k=1}^N n_k \sum_{j=1}^{k-1} \varepsilon_j + \eta \sum_{k=1}^N \frac{n_k^2}{\tau}. \end{aligned}$$

Since $\varepsilon_j \sim N(0, 1)$, the middle term has mean zero. Using $(\sum n_k)^2 = \sum n_k^2 + 2 \sum_k n_k \sum_{j < k} n_j = X^2$:

$$\sum_{k=1}^N n_k \sum_{j=1}^{k-1} n_j = \frac{1}{2} X^2 - \frac{1}{2} \sum_{k=1}^N n_k^2.$$

Taking the continuum limit ($\tau \rightarrow 0$), the $-\frac{\gamma}{2} \sum n_k^2 \tau$ term vanishes (bounded $v(t)$ times $\tau \rightarrow 0$), giving $\mathbb{E}[IS] = \int_0^T \eta v(t)^2 dt + \frac{1}{2} \gamma X^2$.

A.2 Variance of Implementation Shortfall

By Abel summation, $\sum_{k=1}^N n_k \sum_{j=1}^{k-1} \varepsilon_j = \sum_{j=1}^{N-1} x_j \varepsilon_j$. Using independence of ε_j :

$$\text{Var}[IS] = \sigma^2 \tau \sum_{j=1}^N x_j^2.$$

B. Ansatz Decomposition: Financial and Mathematical Justification

B.1 Financial Intuition

At time t , the portfolio holds x shares worth S each. The mark-to-market value is xS . The true value is less because of execution costs (trading fast incurs slippage) and market risk (trading slowly leaves shares exposed to price moves). These frictions depend on remaining inventory x and variance v , but not on the *level* of S .

B.2 Mathematical Justification via the Five-State System

To make the cancellation rigorous, introduce the liquid cash account X_t :

$$dX_t = n_t(S_t - \eta n_t) dt, \quad dx_t = -n_t dt, \quad dS_t = -\gamma n_t dt + \sqrt{v_t} dW_t^S, \quad dv_t = \theta(\omega - v_t) dt + \xi \sqrt{v_t} dW_t^v.$$

The five-dimensional value function $V(t, X, x, S, v) = \sup \mathbb{E}_t[X_T + x_T S_T - \int_t^T \lambda v_s x_s^2 ds]$ satisfies $\partial V / \partial X = 1$ (since X_t appears linearly), so all cross-derivatives involving X vanish in the HJB. Setting $V = X + \mathcal{M}(t, x, S, v)$ and $\mathcal{M} = xS - \Phi(t, x, v)$:

$$\text{Control terms} = nS - \eta n^2 - n(S - \Phi_x) - \gamma nx = -\eta n^2 + n(\Phi_x - \gamma x).$$

The spot price S cancels completely, confirming the ansatz $J = -xS + \Phi(t, x, v)$.

B.3 Brownian Correlation via Cholesky

The drivers dW^S and dW^v are constructed as $W_t^S = B_t^1$, $W_t^v = \rho B_t^1 + \sqrt{1 - \rho^2} B_t^2$ from independent standard Brownians B^1, B^2 . One then verifies $\mathbb{E}[W_t^S W_t^v] = \rho t$ and $dW_t^S dW_t^v = \rho dt$.

C. Euler–Maruyama Discretization and Numerical Stability

C.1 Full Discretization Scheme

With time step $\tau = T/N$ and correlated increments from Appendix D:

$$x_k = x_{k-1} - n_k \tau, \quad S_k = S_{k-1} - \gamma n_k \tau + \sqrt{v_{k-1}} \Delta W_k^S,$$

$$v_k = v_{k-1} + \theta(\omega - v_{k-1})\tau + \xi \sqrt{v_{k-1}} \Delta W_k^v.$$

The log-Euler scheme for price, $S_k = S_{k-1} \exp((\mu u - \frac{1}{2}v_{k-1})\tau + \sqrt{v_{k-1}\tau} Z_k^S)$, avoids negative prices [kloeden1992numerical].

C.2 Feller Condition and Full Truncation

The variance process follows CIR dynamics [cox1985theory]. The *Feller condition* $2\theta\omega \geq \xi^2$ guarantees continuous-time positivity. Euler–Maruyama does not respect this analytically; the full-truncation scheme [glasserman2004monte] floors the variance at zero at each step:

$$v_k \leftarrow \max(v_k, 0).$$

C.3 Path Filtering

Invalid paths (negative price, explosive variance, degenerate IS, incomplete liquidation) are removed symmetrically: if either the AC or Heston evaluation of simulation i is invalid, both are discarded to preserve the paired structure.

Step-size guidance [kloeden1992numerical]: $\tau \leq \frac{1}{2\theta}$ prevents the mean-reversion term from overshooting. For $\theta \approx 2$ this requires $\tau \leq 0.5$; the default $N = 78$ gives $\tau \approx 0.013$, well within stability limits.

D. Cholesky Decomposition for Correlated Shocks

The correlated increments are generated by factoring the correlation matrix:

$$\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} = LL^\top, \quad L = \begin{pmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{pmatrix}.$$

For independent $Z_1, Z_2 \sim \mathcal{N}(0, 1)$:

$$\begin{pmatrix} \Delta W^S \\ \Delta W^v \end{pmatrix} = \sqrt{\tau} L \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} = \sqrt{\tau} \begin{pmatrix} Z_1 \\ \rho Z_1 + \sqrt{1-\rho^2} Z_2 \end{pmatrix}.$$

The leverage correlation $\rho < 0$ in equities means price declines accompany variance increases (“leverage effect”) [black1976studies], causing the correction term to slow liquidation when variance rises.

E. Riccati ODE: Full Derivation and Solution

With $h = -\eta\dot{u}/u$, the Riccati ODE $h_t + \lambda\sigma^2 - h^2/\eta = 0$ becomes $\ddot{u} = \kappa^2 u$ with $u(T) = 0$, solved by $u(t) = \sinh(\kappa(T-t))$. Therefore $h(t) = \eta\kappa \coth(\kappa(T-t))$, which diverges as $t \rightarrow T^-$ —the mathematical signature of the liquidation constraint.

F. Parameter Calibration from LOBSTER Data

F.1 Data Format

The LOBSTER dataset [huang2011lobster] provides tick-level LOB snapshots with L bid/ask levels and a message file recording every order event.

F.2 Volatility and Heston Parameters

Daily volatility is estimated from 5-minute log-returns and scaled by $\sqrt{\text{intervals per day}}$ [CITATION NEEDED: square-root-of-time scaling for volatility, e.g. Campbell, Lo, MacKinlay (1997)].

Heston parameters are estimated from 1-minute rolling realized variance series (scaled to daily units by multiplying by 390 intervals/day). An OLS fit to $\Delta v_t = a + b \cdot v_t$ identifies $\theta = -b/\Delta t$, $\omega = a/(\theta\Delta t)$, and ξ from residual standard deviation after normalizing by $\sqrt{v_t\Delta t}$. The leverage ρ is estimated as the Pearson correlation between 1-minute log-returns and variance changes [CITATION NEEDED: empirical leverage estimation, e.g. Ait-Sahalia–Kimmel (2007)].

F.3 Market Impact Parameters

Temporary impact η is estimated by walking the ask book at simulated trade sizes and fitting $\text{impact} = \eta \cdot X$ through the origin (OLS without intercept) **[CITATION NEEDED: book-walking approach to temporary impact, e.g. Cont–Kukanov–Stoikov (2014)]**.

Permanent impact γ is estimated by regressing the change in mid-price against lagged signed order flow at 5-minute frequency **[CITATION NEEDED: OFR-based permanent impact regression, e.g. Kyle (1985), Almgren et al. (2005)]**.

Risk aversion λ is set by target risk tolerance or the half-liquidation-time criterion **[CITATION NEEDED: standard risk-aversion convention in AC literature]**.

Calibration instability (± 20 – 40% day-to-day) for η and γ is mitigated by exponential smoothing **[CITATION NEEDED: exponential smoothing for parameter estimation, e.g. Holt (1957)]**.

G. Implementation Shortfall Decomposition

$$\text{IS} = \underbrace{\eta \sum_{k=1}^N \frac{n_k^2}{\tau}}_{\text{temporary impact}} + \underbrace{\frac{1}{2} \gamma X^2}_{\text{permanent impact}} + \underbrace{\sum_{k=1}^N x_{k-1} \sqrt{v_{k-1}} \Delta W_k^S}_{\text{price risk realization}}.$$

The mean-variance objective minimizes expected temporary impact plus variance of the price risk term. The permanent impact term is constant (independent of schedule) and drops out.

When v_t is stochastic, the price risk realization has a path-dependent distribution even for a fixed schedule: this additional randomness is precisely what the Heston-adjusted strategy attempts to mitigate.

H. Monte Carlo Simulation Design

H.1 Paired Architecture

Both strategies are evaluated on identical randomness (same (Z_1^k, Z_2^k) sequence per simulation), ensuring $IS_{\text{AC}}^{(i)} - IS_{\text{Heston}}^{(i)}$ reflects only scheduling differences.

H.2 Variance Reduction

Antithetic variates **[glasserman2004monte]**: for each seed, an antithetic path using $(-Z_1^k, -Z_2^k)$ is also simulated and averaged, typically reducing IS estimator variance by 30–50% **[CITATION NEEDED: antithetic variates efficiency gain, e.g. Glasserman (2004) Ch. 4]**.

Note: antithetic variates are described in the mathematical framework but are not currently implemented in the code. The code uses independent per-path seeds.

H.3 Sample Size

Standard error of paired IS mean: $\text{SE} = s_d / \sqrt{M}$. For 95% CI half-width ϵ , the required sample size is $M \geq (1.96 s_d / \epsilon)^2$ **[glasserman2004monte]**. Observed IS standard deviations of 10^7 – 10^8 imply

$M \geq 10,000$ for reliable 1%-level inference.

I. Volatility Regime Partitioning

Realized path volatility $\hat{\sigma}_i = \sqrt{\frac{1}{N} \sum_{k=1}^N v_{k-1}^{(i)}}$ is computed per simulation. Simulations are sorted and split into three equal-size terciles (low/mid/high). A one-sided Wilcoxon signed-rank test within each regime tests whether Heston achieves lower IS than AC in that regime.

The low-volatility-regime statistical significance ($p = 0.0004$, AC outperforms Heston by a fraction of a cent) is discussed in Section I.1.

I.1 Regime Dependence and Model Risk

The regime analysis from the current calibrated runs shows differences of order 10^{-3} (a fraction of a cent per share), statistically significant only in the low-volatility regime and only because 2500 paired observations give very high power even for tiny effects. Several explanations are consistent with the near-zero magnitude:

- (i) **Calibrated ξ is very small.** The LOBSTER variance series for all five equities on the sampled day yields $\xi \approx 0.04$ – 0.05 in daily units. The correction term scales as $O(\xi^2)$ through the product $\xi \cdot h_v = \xi^2 b(t)$, so the absolute magnitude of the scheduling adjustment is $O(\xi^2) \approx 10^{-3}$ —invisible against IS values of order 10^4 – 10^5 .
- (ii) **The Feller condition suppresses ξ .** When calibrated ξ violates the Feller bound $2\theta\omega \geq \xi^2$, the calibrator reduces ξ to maintain numerical stability, further compressing the correction signal.
- (iii) **Higher-order terms are negligible, as is the first-order correction.** The perturbation discards $O(\xi^2)$ terms; with $\xi \approx 0.04$ these are of order 10^{-3} of the first-order term, which is itself negligible.
- (iv) **Filtering bias.** Explosive variance paths are removed symmetrically; the surviving sample may underrepresent high- ξ realizations where the correction would be largest.

The conclusion is that under realistic calibrated market conditions for liquid US equities on a typical trading day, the Heston correction provides no measurable execution advantage. The framework would need either substantially larger ξ (e.g. from a stressed market day) or a longer execution horizon T for the dynamic correction to have economic significance.

J. Statistical Tests: Full Definitions and Hypotheses

J.1 Paired t -Test

Let $d_i = IS_{AC}^{(i)} - IS_{Heston}^{(i)}$. $H_0: \mu_d \leq 0$ vs. $H_1: \mu_d > 0$. Test statistic: $t = \bar{d}/(s_d/\sqrt{M}) \sim t_{M-1}$ under H_0 [lehmann2005testing].

J.2 Wilcoxon Signed-Rank Test

Nonparametric analogue [lehmann2005testing]. Requires only symmetric d_i . Ranks $|d_i|$; computes $W^+ = \sum_{d_i > 0} \text{rank}(|d_i|)$. Under H_0 : $z = (W^+ - M(M+1)/4) / \sqrt{M(M+1)(2M+1)/24} \xrightarrow{d} \mathcal{N}(0, 1)$.

J.3 Levene’s Test for Equality of Variance

H_0 : $\text{Var}[IS_{AC}] = \text{Var}[IS_{\text{Heston}}]$. Define $z_{ij} = |IS_{ij} - \tilde{m}_j|$ (absolute deviation from group median). Apply a two-sample t -test to the z -values. The Brown–Forsythe version (median-centering) is more robust to heavy tails [levене1960robust, brown1974robust].

J.4 Two-Sample KS Test

H_0 : $F_{AC} = F_{\text{Heston}}$. $D_{M,M} = \sup_x |\hat{F}_{AC}(x) - \hat{F}_{\text{Heston}}(x)|$. Under H_0 : $\sqrt{M/2} D_{M,M} \xrightarrow{d} K$ (Kolmogorov distribution) [kolmogorov1933sulla, smirnov1948table].

J.5 CVaR via Bootstrap

$\text{CVaR}_\alpha = \mathbb{E}[IS \mid IS > \text{VaR}_\alpha]$ [rockafellar2000cvar]. Empirical estimate: mean of the top $(1 - \alpha)$ fraction of IS draws. Confidence intervals for the CVaR difference via percentile bootstrap with $B = 10,000$ resamples [efron1993bootstrap].

J.6 Sharpe-Like Reliability Ratio

$\mathcal{R}_{j,d} = \overline{IS}_{j,d} / s_{IS_{j,d}}$. Across-dataset paired t -test: H_0 : $\mathbb{E}[\mathcal{R}_{AC}] \geq \mathbb{E}[\mathcal{R}_{\text{Heston}}]$. Result: $t = 2.876$, $p = 0.006$ [lehmann2005testing].

K. Implementation Details and Engineering Decisions

This appendix summarises the key implementation choices made in the Python codebase, corresponding to the source files `core/AlmgrenChrissModel.py`, `core/MarketEnvironment.py`, `core/Backtester.py`, `core/MonteCarloSimulator.py`, `data/calibrator.py`, `evaluation/comparator.py`, and `evaluation/statistics.py`.

K.1 Time-Scale Convention

All parameters are in units of one trading day ($T = 1$): σ is daily volatility, v_0 and ω are daily variance, and $N = 78$ intervals correspond to one 5-minute bar per interval. Calibrated Heston variance is multiplied by 390 (intervals/day) to convert from per-minute to per-day units before fitting.

K.2 Exact AC Baseline and Corrected Trade Generation

The classic AC schedule is computed from the exact closed-form trajectory $x_k = X \sinh(\kappa(T - t_k)) / \sinh(\kappa T)$. The corrected schedule is computed as a *perturbation on top of the exact baseline*, not by integrating the full ODE with Euler steps. This eliminates a systematic discretization

mismatch that caused the corrected model to underperform AC by $\approx \$1257/\text{path}$ before the fix was applied.

Specifically, the correction chunk at step k is $\delta_k = \text{corr_rate}(t_k, v_k) \cdot \Delta t \cdot x_k^{\text{current}}$, where x_k^{current} is the *running inventory* under the corrected schedule. The inventory scaling is required because the HJB perturbation correction is linear in remaining inventory x_t . After summing, the corrected schedule is renormalized so total sold equals X exactly.

K.3 Quadrature for $b(t)$

The integral defining $b(t)$ is evaluated numerically at each time grid point using `scipy.integrate.quad` (adaptive Gaussian quadrature). A closed-form antiderivative exists but requires case analysis; numerical evaluation is safer and sufficiently fast for $N \leq 390$.

K.4 Parallelisation

Lambda-grid search and dataset sweep use `ProcessPoolExecutor` (process-level parallelism) rather than `ThreadPoolExecutor`, because the Python GIL prevents threads from achieving true CPU parallelism for numerical work [CITATION NEEDED: Python GIL and multiprocessing, e.g. Python documentation]. Worker functions are defined at module level to support cross-platform pickling (required on Windows).

K.5 Random Number Generation

Path generation uses `numpy.random.default_rng` (PCG64 generator), which is thread-safe, statistically superior to the legacy Mersenne Twister, and reproducible from integer seeds [CITATION NEEDED: PCG random number generators, e.g. O’Neill (2014)]. A master RNG generates per-path sub-seeds via `rng.integers`, ensuring reproducibility without correlation between paths.

K.6 Paired Comparison Design

The comparator evaluates both trade schedules against the same Heston path, making the IS difference purely attributable to the scheduling choice. An earlier version used independently drawn paths for each strategy, which invalidated the paired t -test and Wilcoxon test. The fix was identified when the paired test showed zero-variance differences across all 15 datasets, confirming numerical collapse of the corrected schedule to AC.